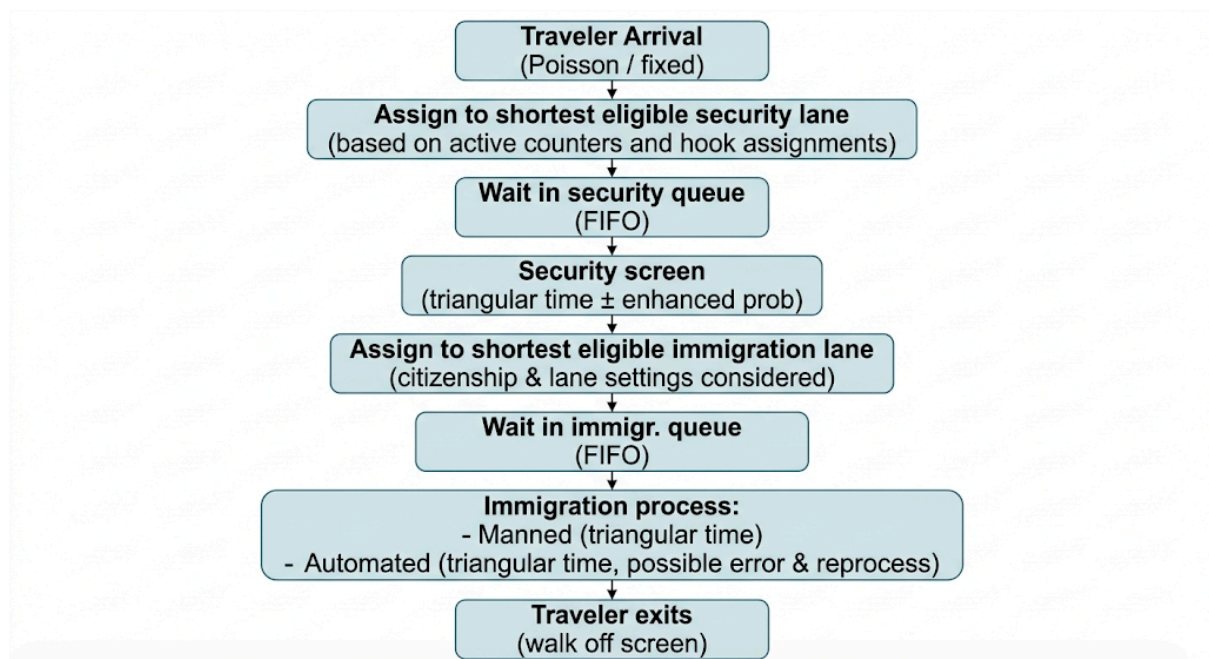


Simulation Design Document

Process Flow

The simulation models a checkpoint with two sequential stages: **Security Screening** followed by **Immigration Clearance**. Travelers spawn at the entrance, queue for an available security lane, get processed, then proceed to an immigration lane (manual or automated), and finally exit.

Flowchart



Key decision points:

- A traveler is assigned to the **shortest eligible lane** among those served by at least one **active counter** (security or immigration).
- For immigration, lane eligibility also depends on the lane's configured rule (**all** / **citizens** / **foreigners**) and the traveler's citizenship.
- Automated counters can only serve citizens and may randomly fail (not bug), causing reprocessing.

System Components

Entity / Resource	Description
Simulation Controller	Central manager: spawns travelers, updates time, processes queues, aggregates metrics, triggers day-end report.
Traveler	Agent with attributes: type (standard / business / family), citizenship, speed, state, timestamps.
Security Lane	4 lanes. Each has a queue of travelers waiting for security. A lane is served if at least one active security counter is hooked to it (or no hooks → nearest counter serves it).
Security Counter	6 fixed positions (Index 0-5). Can be active or inactive (closed). Each can be hooked to one or more security lanes. Displays status (Active, Occupied, Draining, Inactive) and utilization. Draining refers to previously opened counters set to closed, and is now processing the remaining travelers in the queue.
Immigration Lane	4 lanes. Each has a queue and a citizenship rule (All / Citizens only / Foreigners only). Counters can be hooked to lanes to serve them.
Immigration Counter	8 fixed positions. Can be manned (officer) or automated (e-gate). Cycles through Closed → Manned → Automated → Closed on click. Can be hooked to immigration lanes.
Object Pool	Reuses traveler GameObjects to reduce allocation overhead.
UI Manager	Displays real-time metrics, settings panel, day report, traveler logs.

Additional visual components: * **LaneWaypointHolder** – dynamically generates queue positions.

- **LineRenderer** – draws dashed connections between counters and lanes.
- Citizen indicator – a blue passport icon above traveler's head.

Input Parameters

All parameters are configurable via the **Settings Panel** (UI) or through scenario presets.

Category	Parameter	Range / Values
Arrival	Arrival rate (travelers/min)	1 – 20
	Distribution type	Poisson / Fixed
	Citizen percentage	0 – 100%
	Base walking speed (m/s)	1 – 3
	Time-of-day multipliers (8 blocks of 3h)	0.0 – 5.0 each
Traveler Types (auto balanced to 100%)	Standard weight (%)	0 – 100 (auto-balanced)
	Business weight (%)	0 – 100 (auto-balanced)
	Family weight (%)	0 – 100 (auto-balanced)
Security	Number of active lanes	1 – 6
	Processing time – mean (sec)	30 – 120
	Processing time – min (sec)	5 – 120
	Processing time – max (sec)	30 – 240
	Enhanced screening probability	0 – 30%
Immigration	Number of active counters	1 – 8
	Lane assignment rules (per lane)	All / Citizens / Foreigners
	Manual processing – mean (sec)	45 – 180
	Manual – min (sec)	10 – 180
	Manual – max (sec)	45 – 360
	Automated processing – mean (sec)	30 – 120
	Automated – min (sec)	5 – 120
	Automated – max (sec)	30 – 240
	Automated error probability	0 – 50%
	Enhanced screening prob (manual)	0 – 30%

Note: Scenarios (Baseline, Holiday Rush, Optimized, Wet Weather) provide pre-set combinations of these parameters.

Simulation Outputs

The simulation collects and reports the following metrics, displayed in real-time on the top bar and aggregated in a **day report** after 24 simulated hours.

Real-time metrics (updated each frame)

- **Security queue length:** Current total waiting for security (travelers)
- **Immigration queue length:** Current total waiting for immigration (travelers)
- **Security avg. wait:** Average time from arrival until security finishes (minutes)
- **Immigration avg. wait:** Average time from security finish until immigration finishes (minutes)
- **Total avg. wait:** Average total time in system (minutes)
- **Arrival flow (last 60s):** Measured arrival rate (raw and normalised) (travelers/min)
- **Throughput:** Count of exited travelers in last simulated hour (travelers/hour)
- **Staff hours:** Total staff hours accumulated (active counters × time) (hours)

Day report (after 24h) – 3 pages

Page 1 – Overview

- Total travelers processed
- Average throughput (travelers/hour)
- Average system time (minutes)
- Average security wait (minutes)
- Average immigration wait (minutes)
- Peak security queue length
- Peak immigration queue length

Page 2 – Peak hours performance (time multiplier ≥ 2.0)

- Peak hour throughput
- Peak hour average system time
- Peak hour security wait
- Peak hour immigration wait
- Total staff hours

Page 3 – Resource utilization

- For each counter (Security & Immigration):
 - Hours open (active time)
 - Utilization percentage ($\text{busyTime} / \text{activeTime}$)
 - Colour-coded efficiency:
 -  Green: <30%
 -  Yellow: <60%
 -  Orange: <85%
 -  Red: ≥85%

Traveler logs (CSV exportable): Includes traveler's ID, type, citizenship, arrival time, security start/end, immigration start/end, and exit time.

Assumptions

1. **Sequential flow** – Every traveler must complete security **before** entering immigration. No bypassing.
2. **Simulated infinite space** – Travelers have infinite space to queue from each queue waypoint. 1 unit in Unity engine equals to 10 meters (all travelers are 0.17 units in height)
3. **FIFO queues** – Within a lane, travelers are served in the order they arrived.
4. **Processing time distributions** – All processing times follow a **triangular distribution** defined by min, mode (mean), and max values. Enhanced screening adds fixed extra time (45s for security, 60s for manual immigration).
5. **Traveler types** affect walking speed and processing time via multipliers:
 - **Business:** speed +20%, processing time –30%
 - **Family:** speed –20%, processing time +60%
 - **Standard:** speed and time unchanged.
6. **Citizenship** is randomly assigned per traveler based on the **citizenPercentage** parameter.
7. **Automated counters** only serve **citizens** and have a configurable error probability. On error, the traveler is **reprocessed** (same counter, automated time drawn again) with no queue jump.
8. **Enhanced screening** is a random event based on a given probability. It adds extra processing time and shows a warning icon above the traveler.
9. **Counter hooking** – A counter serves **only the lanes it is explicitly hooked to**. If it has no hooks, it serves the geographically nearest lane. When an active counter is closed but still has hooked lanes with queued travelers, it enters **“Draining”** mode and continues serving only those lanes until empty.
10. **Time-of-day multipliers** are applied to the base arrival rate. The day is divided into 8 blocks of 3 hours each.
11. **Simulation duration** – Exactly 24 simulated hours (86,400 seconds). After that, the day report automatically appears and the simulation pauses.
12. **Visual representation** – Travelers are 3D capsules (coloured by type) with a billboarded canvas for citizen indicator (red passport icon). Security counters have a green/red status light; immigration counters show a text label for e-gate vs manned booth.
13. **Object pooling** – Travelers are instantiated once and reused to avoid garbage collection spikes during long runs.